

HYBRID AI-DRIVEN SUPPLY CHAIN ANALYTICS SOLUTION FOR CONSUMER GOODS

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Abstract

The Abstract of the report should be written here, it should provide a short summary of the work encompassing no more than 300 words (roughly). **THE ABSTRACT.TEX FILE IS INSIDE THE INTRO FOLDER**

Acknowledgements

The Acknowledgements section may be used to thank your supervisor, family, research funding bodies, or any other applicable individuals or institutions. The acknowledgements section can be found in `intro/abstract.tex`

Declaration

I confirm that the work contained in this Degree project report has been composed solely by myself and has not been accepted in any previous application for a degree. All sources of information have been specifically acknowledged and all verbatim extracts are distinguished by quotation marks.

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The above student carried out his research project under my supervision.

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Contents

Abstract	ii
Acknowledgements	iii
Declaration	iv
1 Introduction	1
1.1 Chapter Overview	1
1.2 Problem Statement	1
1.3 Research Motivation	1
1.4 Research Gap	2
1.5 Research Questions	2
1.6 Research Aim and Objectives	2
1.7 Significance of the Research	3
1.8 Scope of the Research	3
2 Literature Review	4
2.1 Chapter Overview	4
2.2 Background to the Problem	4
2.3 Theoretical and Conceptual Background	5
2.4 Review of Key Methods and Technologies	9
2.4.1 Data Preprocessing and Feature Engineering	9
2.4.2 Machine Learning Methods	10
2.4.3 Hybrid and Ensemble Methods	11
2.5 Review of Related Studies and Applications	12
2.5.1 Overview of AI-Driven Supply Chain Studies	12
2.5.2 Studies on Data Integration and Preprocessing	13
2.5.3 Machine Learning and Optimization Applications	13
2.6 Identified Research Gaps and Justification	14

2.7	Chapter Summary	15
3	Design	16
3.1	Conclusions	16
4	Implementation	17
4.1	Conclusions	17
5	Evaluation & Testing	18
5.1	Conclusions	18
6	Conclusion	19
6.1	Conclusions	19
6.2	Future Work	19
7	Using $\LaTeX$	20
7.1	Structure of this Template	20
7.2	Using Figures	21
7.3	Referencing and Hyperlinking within the Document	22
7.4	Citing Bibliographic References	23
7.4.1	The Harvard Bibliography Style	23
7.4.2	The Vancouver Bibliography Style	24
7.4.3	Compiling a BibTeX Database	24
7.5	Working with Tables	24
7.6	Inserting an Algorithm	25
7.7	Inserting Program Code Samples	25
7.8	Working with Maths	26
7.9	Required Software	26
7.10	Working with Quotes	27
7.11	Further Information	27
A	Project Specification	34
A.1	Functional Requirements	34
A.2	Non-Functional Requirements	34
B	Project Management	35
C	Another Appendix	36
D	Presentation Slides	39
E	Project Log	40

List of Tables

7.1	Table Caption	24
C.1	A Short Caption for the table	37

List of Figures

7.1	Mobile Phone Suppliers, Market Share 2006 (Millions of Units Shipped)	21
7.2	Caption for Figure	22
7.3	Caption for Figure	22
7.4	Caption for Bitmap Image Example	22
C.1	A Sideways Figure	38

List of Algorithms

1	A Sample Algorithm	25
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Listings

7.1	The Caption for the Code Listing	25
7.2	Sample Program Code Listing	26

Chapter 1

Introduction

1.1 Chapter Overview

A number of pages about the background of the project.

1.2 Problem Statement

Provide a list of all the chapters within the thesis and a brief summary of the content.

1.3 Research Motivation

The increasing complexity of global supply chains and consumer demand volatility have intensified the need for intelligent, adaptive, and transparent analytics systems in the consumer goods sector. Recent studies reveal that AI-driven forecasting can reduce prediction errors by 15–20%, resulting in improved inventory turnover and cost efficiency (Khedr and S [2024a](#)). Nevertheless, all those advantages do not always mean that the introduction of AI solutions will be welcomed by organizations because such solutions are not explainable, do not align well with business goals, and they do not fit the cost structure. The conventional forecasting mechanisms consider over and under-forecasting as an equal offense and ignore their unique financial and service-level consequences (Ledro et al. [2025](#)). In addition, the current sustainability requirements and the trends of the circular economy demand reverse logistics returns, recycling, and remanufacturing modeling that are still mostly nonexistent in the majority of analytics models. The rationale behind this research is the necessity to develop a hybrid, interpretable, business-oriented AI that will theorize the gap between data science and strategic supply chain management. This research will contribute to

generating practical findings, which will not only provide a better forecast but also a higher level of resilience, profitability, and sustainability in consumer goods supply chains by combining causal explainability, extraneous data streams, and cost-sensitive optimization.

1.4 Research Gap

1.5 Research Questions

This research is informed by a research question which is central and has a series of secondary questions which examine various facets of the problem. The questions below are designed to guide the research and offer an orderly manner of answering research objectives.

Primary Research Question

- **RQ1:** How can a hybrid AI-driven analytics framework enhance the accuracy, transparency, and cost-effectiveness of supply chain forecasting in consumer goods companies?

Secondary Research Questions

- **RQ2:** What combination of internal and external data sources (e.g., sales history, macroeconomic indicators, social sentiment) most effectively improves forecast performance in consumer goods supply chains?
- **RQ3:** How can explainable AI (XAI) techniques be integrated into forecasting models to enhance planner trust and decision transparency?
- **RQ4:** In what ways can cost-sensitive and asymmetric optimization functions better align AI-based forecasts with business performance goals such as service level, inventory turnover, and sustainability outcomes?

1.6 Research Aim and Objectives

The objectives of the research are as follows and outline the desired direction and quantifiable goals of the research. They will act as a guide on how the research will be carried out and provide some form of consistency on the purpose and the anticipated outcomes of the research.

Primary Objective

- **RO1:** To design and evaluate a hybrid AI-driven supply chain analytics framework that integrates interpretability, real-time external data, and cost-sensitive optimization to enhance forecast accuracy and operational resilience in consumer goods companies.

Secondary Objectives

- **RO2:** To identify and analyze the impact of integrating diverse data sources internal (sales, logistics) and external (economic, sentiment, weather) on demand accuracy.
- **RO3:** To develop and test explainability mechanisms (e.g., SHAP, LIME) that make AI-based forecasting models transparent and trustworthy for supply chain planners.
- **RO4:** To incorporate asymmetric cost-aware optimization functions within forecasting and replenishment policies, aligning predictive insights with business outcomes such as reduced stockouts, improved inventory turnover, and enhanced sustainability.

1.7 Significance of the Research

1.8 Scope of the Research

Chapter 2

Literature Review

2.1 Chapter Overview

This chapter aims at critically discussing the scholarly literature on AI-based supply chain analytics in the context of consumer goods, with the aim of giving a holistic view of theoretical basis, technological inventions, and research outcomes upon which this study draws upon. Integrating studies on the use of machine learning, as well as hybrid intelligence, and the use of data analytics in supply chain management, this review proposes the novelty of research gaps in the literature, emphasizing methods, and locating the study in the context of the larger academic discussion of digital transformation in logistics. The three thematic dimensions (technological innovation, organizational transformation, and performance outcomes) used to organize the chapter enables to have a clear idea of how the AI-human-collaboration can contribute to improving supply chain visibility, flexibility, and sustainability. By means of this synthesis, the review provides the theoretical and practical premises to the development of a hybrid AI-based analytics tool, in which the needs of global consumer goods supply chains will be adjusted to the new requirements, and key gaps in the concept of ethical AI implementation, data integration challenges, and emerging market uses, which can justify the necessity to conduct additional research, are identified.

2.2 Background to the Problem

Consumer goods companies such as Unilever and Nestle supply chain management are being increasingly focused on proper demand forecasting and nimble decision-making to control inventory and costs (Luo [2024](#)). The conventional time-series models, like ARIMA and Holt-Winters, do not perform well when the demand is volatile

and unpredictable disruptions occur. The most recent developments in artificial intelligence (AI) and machine learning (ML) such as LSTM and XGBoost have increased the accuracy of the forecast by 15 to 20 percent, and this is now applicable to inventory turnover, stockouts, and cost reduction (Chowdhury, Paul, and Rozony 2025a). Nevertheless, there are still difficulties: AI models are generally not transparent enough to give the planners confidence, and most of them are backward-looking, which means a significant dependence on past sales and a lack of support to real-time drivers of the market, including macroeconomic or social indicators.

Also, such standard measures as MAPE do not take into account asymmetric business costs of over- and under-forecasting. Forecasting is also not integrated with prescriptive decision systems and does not consider reverse logistics which is a key aspect in e-commerce and sustainability. To overcome these shortcomings, this proposal presents a hybrid, interpretable, and cost-sensitive model of analytics using statistical, signal-processing, and ML methods. It involves external information in real time, explainability procedures, asymmetric maximization of business goals and reverse-flow predictions. It aims to offer strong, transparent and practical forecasts that complement efficiencies, cost-effectiveness, and sustainability in consumer goods supply chains.

2.3 Theoretical and Conceptual Background

The combination of Data Science and Artificial Intelligence (AI) in supply chain analytics has altered the old, linear systems of operation to smart, self learning and adaptable networks. The AI provides predictive, informative, and strategic capabilities in the consumer goods industry, where the demand is volatile, the product life cycle is short, and logistics are complex. This section will create the theoretical base and conceptual knowledge in which the study will be based. It describes the role of AI, big data, and hybrid analytics in creating more resilient, smart, and sustainable supply chains, as well as specifying key terms, ideas, and models used to formulate the intelligent supply chain in the modern context.

Theoretically speaking, a number of frameworks describe the ways in which AI-based analytics enhances the performance of the supply chain and competitive edge. According to the Resource-Based View (RBV) theory (Barney 1991), firms can experience a sustainable advantage in case they bear resources that are unique, valuable and inimitable (Jackson et al. 2024). These key resources in the contemporary world are digital assets, including information infrastructure, artificial intelligence, and analysis capabilities (Mailani et al. 2024). Using AI to convert gigantic quantities of raw data into predictive and prescriptive data, consumer goods companies will have successfully turned digital capability into a strategic asset that aids decision-making and continuous

improvement (Lopes de Sousa Jabbour et al. [2025](#); Jauhar et al. [2024](#); Aghaei et al. [2025](#)).

This is further complemented by the Dynamic Capabilities Theory (Teece [2007](#)) which provides an explanation of how the firms adapt to the volatile environments in response to the capability to sense opportunities and seize them and reorganize internal processes. These dynamic capabilities are enhanced by AI and machine learning since it enables supply chain managers to identify the changes in the market, predict disruption, and real-time adjustment of the decisions (Toth, Goga, and Boscoianu [2025](#)). An example is AI-based demand forecasting which assists companies to respond to seasonal changes in sales, evolving customer preferences, or delays in supply to continue with the service even when uncertain (Ahn et al. [2024](#); Kumar et al. [2024](#)).

Also, there is the Sociotechnical Systems Theory (Bostrom and Heinen [1977](#)), which puts an emphasis on harmonizing technology with humans. The effectiveness of the AI integration process is not determined only by the technology itself but also by the level at which the technology can collaborate with the human expertise. This view is core to hybrid AI-based supply chains, where AI analytics is supplemented by human judgment to deal with ambiguous, ethical or context-specific decisions. Instead of substituting human beings, hybrid systems enhance human knowledge- a kind of balancing process where technology can be applied in analyzing data, whereas human beings can monitor, process, and authenticate decisions (Vann Yaroson, Abadie, and Roux [2025](#)). This principle of augmented intelligence creates responsible and efficient decision-making in place of a complicated supply chain in the world.

In this context, it is essential to determine the main concepts in order to comprehend AI and Data Science. Artificial Intelligence (AI) is offered as a term to describe computer systems that can replicate human cognitive processes, including reasoning, learning, and problem-solving. In the supply chain, AI is represented by technologies, such as Machine Learning (ML), Deep Learning (DL), Natural language Processing (NLP), and Computer Vision, each of which has a specific use of analysis. Examples of machine learning are models that, through historical data, learn to make predictions and become more accurate as time progresses without being coded. Supervised ML models are used in supply chain analytics to predict product demand or pricing, whereas unsupervised clustering methods can be used to group suppliers or customer groups together because of similar behavior patterns (Khedr and S [2024b](#)). Deep learning, which is a subdivision of ML, operates on large and unstructured data, e.g., sensor data, customer reviews, or image data, and allows identifying defects and making dynamic predictions in logistics (Hosseinnia Shavaki and Ebrahimi Ghahnavieh [2023](#); Riesen [2025](#)).

Other necessary concepts are predictive analytics and prescriptive analytics. Predictive analytics involves the application of algorithms to forecast events ahead such as sales volumes, supplier delays or stockouts whereas prescriptive analytics goes a step further by suggesting the best decisions to make in different circumstances e.g. when to replenish inventory or reroute deliveries. These processes are based on the Big Data Analytics (BDA) that means gathering the information provided by various sources, such as enterprise systems, IoT sensors, and market data, into one analytical platform. Nevertheless, (Kache and Seuring 2017) observe that the quality of data, its governance, and interoperability are key to the success of AI. Even the best algorithms are unlikely to give accurate or biased results without trustworthy real-time information.

The other important concept is the concept of Hybrid Intelligence which defines the cooperation of human thought and artificial intelligence. Rather than being completely automated, hybrid systems use AI-based computation and human-based context reasoning. Hybrid intelligence in the consumer goods business will create a balance between efficiency and responsibility in the industry where ethical and sustainability factors play a role in supply decisions (Vann Yaroson, Abadie, and Roux 2025). This practice matches the current organizational trends of focusing on transparency, human control, and ethical governance of AI implementation.

To conceptualize these notions, there are multiple frameworks that can be used to describe the adoption of AI into the supply chain system. The Supply Chain Operations Reference (SCOR) Model is one of the oldest to organize the supply chains according to the five fundamental dimensions, Plan, Source, Make, Deliver, and Return (DERFOUFI and BENRREZZOUQ n.d.). AI improves all aspects of it by using automation and analytics: predictive planning is more precise, AI-based procurement is more efficient at choosing suppliers, robotics make the manufacturing process more efficient, and real-time tracking makes the distribution process smoother. The integration of AI into the SCOR model will turn passive processes into active, data-oriented systems that learn and become adapted at all costs.

The other applicable model is the AI-Augmented Decision-Making Model that has three layers of analytics: the descriptive, predictive, and prescriptive. At the descriptive level, AI systems represent the existing performance in form of dashboards and real-time monitoring solutions. The predictive layer uses machine learning to predict disruption or demand variations whereas the prescriptive layer gives practical advice including rerouting transport or modifying production timelines (Herath et al. 2023). These suggestions are verified by managers so that decisions are not made outside of organizational context and ethics. The approach to hybrid intelligence is based on this framework, which is known as the human-in-the-loop approach.

The DIKW Hierarchy (Data-Information-Knowledge-Wisdom) is yet another description of the process of knowledge transformation in systems that rely on AI. Information derived through different sources turns into data once processed and arranged; AI can turn this into knowledge by recognizing latent patterns and correlations; human specialists can turn this knowledge into wisdom turning insights into actionable business strategies. In a hybrid supply chain, AI can speed up the chain of command of any supply chain, whereas the human factor provides the wisdom component with a sense of context (Deepu and Ravi [2021](#)).

The new framework in digital supply chains is the Dynamic Digital Twin Model, in which AI-based simulations generate virtual images of the physical supply chain. These digital twins are always informed by the real-world data enabling managers to simulate various situations like supply disruptions or sudden demand spikes and then execute the decisions in the real-time (Espinosa-Jaramillo [2024](#)). Digital twins have the ability to simulate the impact of alteration of a production scenario on inventory, transportation, and energy consumption in consumer goods production, and thus make decisions out of data and preemptively (Barykin et al. [2020](#); Abideen et al. [2021](#)).

Data-driven management also has a number of principles that govern the application of AI in supply chains. The interoperability and integration of data are building blocks because any AI system cannot work without a integrated data stream between procurement, production, logistics and sales. The clarity and explicability are also of great essence; the AI models should be readable in order to be understood by the human managers, to be validated and, in case, override automated decisions. This increases confidence and eliminates threats of algorithm bias. Scalability and real-time responsiveness are a plus in that AI systems can process large amounts of data and deliver insights in real-time, which is needed by the fast-moving consumer goods (FMCG) industry (Ghodake et al. [2024](#)). Finally, responsible use of technology based on the ethical and sustainable frameworks of AI is necessary to be fair, inclusive, and cause minimal damage to the environment (Analytics [2025](#)). The ethical AI practices enable corporate objectives in the Walmart style eco-reduction, fair labor, and green logistics.

The idea of an AI-based supply chain analytics hybrid concept could be represented in the shape of a multi-layered ecosystem by combining these theories and frameworks. The data acquisition layer is the first layer and gathers both structured and unstructured data on suppliers, production systems and markets. The second layer- AI analytics, makes use of all these datasets through machine learning to make predictive insights. The third layer -human-AI collaboration enables the managerial interpretation and AI ethical validation of the outputs. The fourth layer is a decision execution

which realizes data-driven activities in supply chain functions, and the fifth layer is feedback and learning which captures performance data to re-train AI models to improve subsequent decisions. This continuous learning so ascertained is a pointer to the Dynamic Capabilities approach because it senses change, seizes opportunity and reinvents operations to ensure resistance to change and competition (Herath et al. 2023).

2.4 Review of Key Methods and Technologies

The need to implement Artificial Intelligence (AI), machine learning (ML), and advanced analytics has been predetermined by the growing number of supply chains of consumer goods and their complexity and data intensity (Samuels 2025; Sharma et al. 2022). This section is a critical analysis of the major methods, algorithms, and tools that are used in hybrid AI-driven supply chain analytics. It talks about preprocessing methods, methods of evaluation of the model, and contrasts the merits and demerits of various algorithmic and analytical methods in the context of predictive forecasting, optimization, and decision automation.

2.4.1 Data Preprocessing and Feature Engineering

The data preprocessing is an important aspect of supply chain analytics because it helps to ensure that raw and heterogeneous data may be transformed into structured and trustworthy inputs of AI models. The enterprise systems and IoT devices, as well as social media, provide supply chains with large amounts of transactional, sensor, and unstructured textual data (Sodhi and Tang 2021). Best preprocessing ensures the accuracy of model and avoids overfitting or bias. Data cleaning, normalization, feature extraction, and dimensionality reduction are some of the key techniques in this process (Min, Zacharia, and Smith 2022).

Data cleaning in the consumer goods environment eliminates duplicate data, absent values, anomalies in order information and logistics files (Guo et al. 2023). Normalization makes the various measurement scales comparable whereas feature engineering isolates important attributes like lead time variability, shelf- life or supplier reliability that enhance better model learning. Principal Component Analysis (PCA), as well as t-Distributed Stochastic Neighbor Embedding (t-SNE) is commonly used to lower the dimensions and define the most significant variables with their impact on demand and delivery (Jahin et al. 2024).

Time-series decomposition is another preprocessing method that is required to isolate seasonal and trend elements of the sales or inventory data, and to enable predictive

models to be more understandable. Moreover, within the hybrid AI, data fusion is used in order to integrate structured ERP data with real-time IoT sensor measurements (e.g., temperature, humidity, vehicle speed) and unstructured social data (consumer reviews, social trends) to increase the level of contextual intelligence (Ivanov, Tsipoulanidis, and Schönberger 2025). Correct preprocessing therefore forms a solid data base that enhances the efficiency of the following machine learning and optimization models.

2.4.2 Machine Learning Methods

Machine learning algorithms are being massively employed in supplier chain forecasting, demand sensing, and anomaly detection of consumer goods. Supervised models of learning are generally employed in case of labelled data, whereas unsupervised and reinforcement learning are employed in clustering and adaptive decision-making.

Linear Regression and Multiple Regression are some of the basic models that are used in identifying the relationship that exists between demand and any force that influences the demand like price, promotion or seasonality. These models however are linear and independent of variables and therefore, they are less predictive to complex nonlinear situations. High-level algorithms like Random Forests (RF) and Gradient Boosting Machines (GBM) overcome this shortcoming by combining numerous decision trees in order to achieve generalization and nonlinearity (Breiman 2001). Retail demand forecasting literature indicates that RF models are better than the conventional regression models in their accuracy and robustness, especially when high-dimensional noisy data are to be considered (Mitra et al. 2022).

Another popular method that is applied in classification and regression tasks is Support Vector Machines (SVMs). They are useful in the identification of nonlinear patterns through the use of kernel functions. SVMs may categorize the suppliers or the shipments as a high-risk or a low-risk group, according to the multidimensional variables like geopolitical indicators, shipment delays, and financial variables (L. Zhao, Qi, and Zhu 2021). SVMs however necessitate a lot of hyperparameter optimization and can be computationally inefficient with large volumes of data.

Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks of Artificial Neural Networks (ANNs) and Deep Learning (DL) have become the focus of time-series prediction and dynamic inventory management. The RNNs are able to capture sequential dependencies and can therefore be used to predict consumer demand patterns or shipment delays. LSTMs specifically are able to retain long-term dependency as well as cope with anomalous changes, which provides high predictive rates of demand forecasting in FMCG (Fast-Moving Consumer Goods) markets (Godoy, Arias, and Franco 2023). Nevertheless, they consume high computational resources and

large volumes of labelled data making them less useful in small-scale operations (Na 2024; Liu and Vakharia 2024).

Recent research also notes that Reinforcement Learning (RL) can be implemented in the supply chain to make adaptive decisions. The RL agents are taught the best actions (e.g. replenishment quantity, transport scheduling) using trial and feedback, as a simulation of dynamic and uncertain environments. Some uses are optimization of routes and inventory allocation, in which RL continuously improves methods to reduce costs and delays (Chowdhury, Paul, and Rozony 2025b). Still, inferential quality and time to convergence are the most important issues with RL models.

2.4.3 Hybrid and Ensemble Methods

Hybrid analytics techniques involve the use of more than one algorithm or machine learning with established optimization and simulation techniques to ensure higher accuracy and flexibility. Ensemble methods like Bagging, Boosting and Stacking combine the result of multiple base models in order to generate a more robust meta-model. An example is that Extreme Gradient Boosting (XGBoost) and Light Gradient Boosting machine (LightGBM) have been shown to have a higher predictive accuracy in terms of sales and inventory than single models (Ke et al. 2017).

The second way of hybridity is the use of AI models together with Operations Research (OR) optimization tools. Indicatively, predictive models have the capability to determine the future demand and optimization tools such as Mixed Integer Linear Programming (MILP) or Genetic Algorithms (GA) can select the most appropriate inventory and transport assignments to make based on the predicted demand. Such integration aids in predictive as well as prescriptive decisions in supply chains (Wamba, Dubey, and Gunasekaran 2022).

The other hybrid method is a combination of simulation modelling (e.g., system dynamics, agent-based modelling) and AI algorithms. Such systems have the ability to emulate the occurrences of disruptions, like port delays or suppliers' failure and apply AI to optimize responses in real time. As it was proven by (Ivanov, Tsipoulanis, and Schönberger 2025), digital twins simulations in combination with ML predict supply chain resilience to diverse stressors. The hybrid models, however, have complexity in data integration, model transparency and cost of computation which may pose an obstacle to scalability.

2.5 Review of Related Studies and Applications

The application of Artificial Intelligence (AI) and analytics in supply chain management has been much discussed within the various sectors and especially consumer goods and logistics. In this section, the reviews are critical of key empirical studies, industry applications, and research trends regarding the AI-driven and hybrid analytics systems in global supply chains. The review will focus on establishing generalized findings and issues and implications to the creation of a hybrid AI-based supply chain analytics solution in the consumer goods.

2.5.1 Overview of AI-Driven Supply Chain Studies

During the last ten years, the scholarly and commercial interest in AI-enabled supply chain analytics has increased exponentially. The studies show that AI enhances visibility of supply chains, prediction, and resiliency. In one example, (Ivanov, Tsipoulanidis, and Schönberger 2025) have revealed that machine learning and predictive analytics enable minimizing uncertainty by enhancing the global logistics system responsiveness. Equally, (Wamba, Dubey, and Gunasekaran 2022) demonstrated that AI leads to a more efficient decision-making process by converting descriptive information into a course of action.

The implementation of the Big Data Analytics (BDA) and predictive modelling can also be supported empirically as enablers of supply chain agility. (Kache and Seuring 2017) argue that the data integration among various nodes (suppliers, warehouses, and customers) will help in real-time synchronization. This enables companies like Walmart and Procter and Gamble to lower lead times and enhance supply demand equilibrium. Based on case studies, companies that implement AI forecasting models (e.g., Unilever predictive demand model) record up to 20 percentage points increase in inventory turnover and up to 30 percent decline in the amount of unused stocks (Min, Zacharia, and Smith 2022).

Besides, hybrid systems combining AI with more traditional methods of optimization or simulation are more adaptable. As noted by Ivanov (2023)(Ivanov 2023), disruptions in physical supply chains can be simulated with the help of digital twins, which are computer-based replicas of real-life supply chains, and the corrective actions should be assessed, and this greatly enhances supply chain resilience. Due to the implementation of digital twins and predictive modelling, it is possible to conduct the analysis under the conditions of uncertainty in the form of a “what-if analysis with the assistance of which the managers can predict how geopolitical instability or disruption caused by the pandemic will affect product flow.

2.5.2 Studies on Data Integration and Preprocessing

Studies note that data diversity is one of the critical hurdles of implementation AI to supply chains. Research by (Jahin et al. 2024) emphasizes that predictive performance directly relies on data quality. They found preprocessing tools such as feature extraction, normalization and dimensionality reduction to be essential in model accuracy. Unstructured data in the form of sensor readings, reviews of texts, and social signals, among others, has to be transformed into structured forms in AI-based supply chain forecasting.

The concept of data fusion, referring to the combination of real-time IoT data with ERP and sales systems, has become an essential approach to the attainment of end-to-end visibility in the consumer goods logistics. According to the Future of SCM report, blockchain-based data structures are implemented by other large corporations like IBM and Maersk to create links between shipping documents, container tracking, and custom approvals to minimize delays, and fraud (Rejeb, Keogh, and Treiblmaier 2019). In a similar manner, the partnership between Walmart and IBM in blockchain traceability enhanced food safety since it cut down on the time food products are tracked by seven days to only 2.2 seconds. These practical examples are confirmation that data integration and secure transparency is not just a technical necessity but also a strategic facilitator of trust in consumers and regulatory compliance.

2.5.3 Machine Learning and Optimization Applications

The application of machine learning techniques has been widespread to the important supply chain operations like demand forecasting, procurement and optimization of transportation. A research conducted by (Thejovathi n.d.; Kang 2023) revealed that having nonlinear relationship capabilities, Random Forest (RF) and Gradient Boosting algorithms are more efficient in predicting the sales of FMCG than linear regression. Deep learning networks like Long Short-Term Memory (LM) networks have been effectively used to forecast short term demand changes and inventory. (Godoy, Arias, and Franco 2023) documented that LSTM-based prediction increased the accuracy of the prediction by 15-25 percent in comparison with ARIMA prediction models in fast-moving consumer goods.

Adaptive logistics studies have been studied with a focus on optimization via Reinforcement Learning (RL). It allows systems to learn the best decisions with simulation and feedback that is made possible by RL. The study by (Chowdhury, Paul, and Rozony 2025b) showed the ability of RL to enhance routing and fleet usage and reduce the transport expenses by up to 18 percent in experimental logistics. Nevertheless, it is not widely used in practice because of the high cost of computations

and constant retraining.

2.6 Identified Research Gaps and Justification

Even though recent research proves that recent scholarship makes a considerable leap in implementing AI and data-driven analytics towards supply chain management, there are still multiple important gaps that support the rationale behind the current research. To begin with, the bulk of existing research has been done on individual-functional or limited uses of AI, e.g., demand forecasting, routing optimization, or inventory control, and not a unified, hybrid framework that brings together predictive, prescriptive, and human-validated analytics throughout the entire supply-chain lifecycle. Both (Ivanov, Tsipoulanidis, and Schönberger 2025) and (Wamba, Dubey, and Gunasekaran 2022) admit that the following step of development is the cross-functional integration, but there is not much empirical evidence on how hybrid architectures can be implemented in consumer-goods settings.

Second, there is a gap in data-governance and interoperability, which the literature shows has persisted. Researchers like (Kache and Seuring 2017) can point to data fragmentation, the lack of standards, and weak transparency as the factors that diminish the reliability of algorithms. Although blockchain and IoT-based traceability systems demonstrated their potential, there is a lack of studies on how the technologies can be integrated with AI analytics to formulations of unified and interoperable ecosystems enabling trust, data protection, and ethical standards.

Third, the human-AI co-operation aspect, which is the focus of hybrid intelligence, has not been well investigated in consumer-goods supply chains. Current models tend to presuppose complete automation, neglecting the socio-technical interaction that is necessary to make contextual decisions, engage in ethical monitoring and explain them (Arrieta et al. 2020). This creates a gap in knowledge as how to incorporate human judgment in AI-influenced decisions to find the balance between efficiency and accountability.

Fourth, there is a lack of theorisation of a sustainability connection between the AI-enabled analytics and the environmental or social context. Majority of AI research is about efficiency or cost reduction, but few evaluate the way predictive analytics can actually assist in ESG measures like carbon-footprint minimisation, reduction of waste, or responsible sourcing (Nogueira, Assis Rangel, and Shimoda 2021). It is right to fill this gap in line with the increasing international focus on green and ethical supply-chain practices.

Lastly, there exists a contextual gap in the manifestation of emerging-market supply

chain. North American or European contexts are the basis of much of the empirical evidence, and there is not much understanding of how hybrid AI systems might be scaled to environments with limited data and resources like infrastructure, which are common in consumer-goods markets in Asia-Pacific.

This paper bridges these gaps by proposing a hybrid AI-based supply-chain analytics system with predictive and prescriptive intelligence, transparency with blockchain capabilities, and the human-in-the-loop decision-making process. The new knowledge brought by the proposed model is that it would show how data science, sustainability metrics, and managerial judgment can be integrated into a comprehensive, ethically responsible analytical ecosystem. By so doing, it fosters theoretical knowledge as well as practical implementation measures of next-generation supply chain of the consumer-goods industry.

2.7 Chapter Summary

The chapter has integrated a large body of both theoretical and empirical research relating to AI-driven and hybrid supply-chain analytics. It laid the formal ground with Resource-Based View, Dynamic Capabilities, and Socio-technical Systems theories, and examined a very broad spectrum of analysis mechanisms - including machine learning, deep learning, optimization and hybrid combinations. Related studies and industry applications were also discussed in the chapter, with successes in accuracy of the forecast, data integration, and digital-twin simulation and lack of success through issues of data quality, interpretability, and ethical governance.

Based on these observations, a number of gaps in research were identified with regards to integration, interoperability, human collaboration, sustainability, and applicability in emerging markets. The results were used to construct the hybrid AI-oriented framework used in the current study, which intends to provide consumer-goods supply-chain with an adaptable, transparent, and sustainable analytics solution. The following chapter will hence map the research methodology, covering the research design, methods of data-collection, analysis procedures, and strategies of validation employed in order to test empirically this proposed model.

Chapter 3

Design

This chapter examines the design of the project.

3.1 Conclusions

The main conclusions for this chapter.

Chapter 4

Implementation

This chapter examines the implementation of the project.

4.1 Conclusions

The main conclusions for this chapter.

Chapter 5

Evaluation & Testing

This chapter evaluates the overall project and provides results of tests carried out.

5.1 Conclusions

The main conclusions for this chapter.

Chapter 6

Conclusion

This chapter summarises the main outcomes and conclusions resulting from this body of work.

6.1 Conclusions

The main conclusions that may be drawn from the body of work.

6.2 Future Work

Further development that could be carried out in the future.

Chapter 7

Using L^AT_EX

There are several reasons why one should prefer L^AT_EX to a *what you see is what you get*(WYSIWYG) word processor like Microsoft Word: *portability, lightness, security* are just a few of them (not to mention that L^AT_EX is free!). There is still a further reason that should definitely convince you to abandon MS Word for the development of a dissertation: you will never be able to produce professionally typeset and well-structured documents using most standard WYSIWYG tools.

L^AT_EX is a free typesetting system that allows you to focus on content without bothering about the layout: the software takes care of the actual typesetting, structuring and page formatting, producing documents of astonishing elegance.

7.1 Structure of this Template

The file *thesis.tex* in the root of the directory (ThesisTemplate) is the main file of this template. This is the file that must be compiled to create the document. The *thesis.tex* document contains a lot of configuration settings. The only elements that require editing are details such as the title of the report, authors name and so forth. The only further addition to the file is to use the `\include` statement to include additional chapters into the report. One may also comment out the `\include` statements using the percentage sign (%) to develop the report on a chapter by chapter basis. The BibTeX database *thesis.bib* is also included within the root. All the actual content of the report is divided up into directories each with a *.tex* file containing the chapter content.

7.2 Using Figures

One can insert graphic elements using L^AT_EX in a number of ways. Vector based imagery such as diagrams saved to the pdf format may use the `\includegraphics` command with the optional *viewport* attribute to specify a precise area of the graphic to be included. Figures should also include a Caption and a Label for referencing.

When inserting a Figure (Figure 7.1) one uses the `\begin{figure}` and `\end{figure}` commands. The image presented is a vector graphic in the form of a pdf file. When working with such files it is usually necessary to include the optional *viewport* attribute to designate the specific area in which to focus. The first pair of coordinates (x & y) designate the pixel location of the lower left corner. The second pair identify the upper right hand corner. Modification of these coordinates allows one to focus in upon a particular area of interest within the vector image. The optional attribute [H] when beginning a Figure inserts the graphic element at the specified location. Other options such as [htb] (here, top, bottom) will place the graphic in the most suitable place that L^AT_EX can find. This however can have a negative impact on memory allocation if a large number of images are to be found within the document.

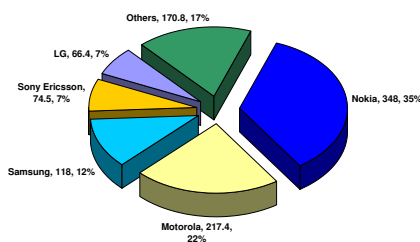


Figure 7.1: Mobile Phone Suppliers, Market Share 2006 (Millions of Units Shipped)

One may use the *minipage* command when inserting two figures to span across the page. This allows for the subdividing of the page into a number of columns of specified width. Note the pie-charts presented here (Figure 7.2 & 7.3) are a bit small for viewing as printed matter. Zoom in on them using a pdf reader to see the advantage of printing diagrams (vector graphics) in pdf format.

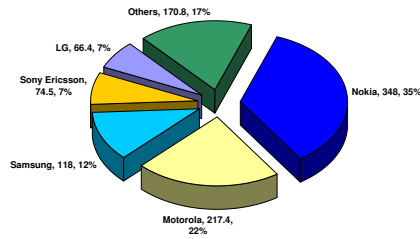


Figure 7.2: Caption for Figure

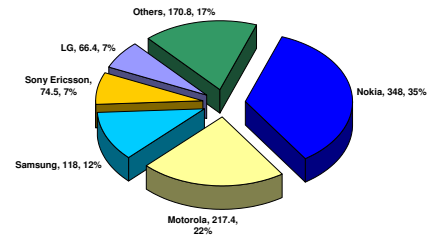


Figure 7.3: Caption for Figure

The example below (Figure 7.4) demonstrates the insertion of a bitmap image. One can see that the extension for the image file isn't specified, as this template is setup to automatically search for .jpg, .png, .gif and .pdf images. The size of the displayed image within the document can be varied by adjusting the height and width attributes. To rotate an image 90 degrees an optional attribute can be added, for example [width=.4\linewidth, angle=90].

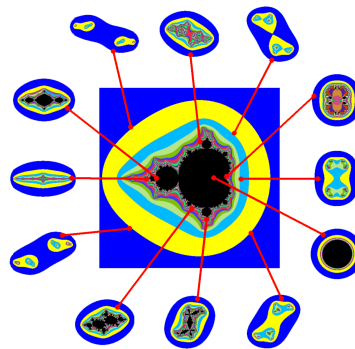


Figure 7.4: Caption for Bitmap Image Example

7.3 Referencing and Hyperlinking within the Document

To refer to another part of the document one must use a combination of the `\ref` and `\label` commands. The label is a unique identifier, therefore when working with large documents it helps to give references meaningful names. Examples of this includes prefixing Table references with *tab:*, figures with *fig:*, chapters with *ch:*. In very large documents it may also be useful to add an additional level of prefixing to represent the chapter the label is in. In this example chapter the tables and figures have the additional prefix of *using* to represent the *using_{Latex}* chapter. The tilde (`~`) is used to ensure that a reference remains as a single object. All instances of `\ref` should be preceded with the tilde.

7.4 Citing Bibliographic References

Bibliographic references are stored in a database (.bib file extension), this contains a list of articles, proceedings, books, thesis and so forth. Each type of publication has a number of required fields such as a unique identifier, author and title. To cite a references within the main body text one must use the `\cite` command as in the following examples. Einstein [1956](#) for example is well known for his work on Brownian Movement. The SETI@Home project Berkeley University [2010](#) is an example of a webpage citation. One can also work with articles Akenine-Möller and Ström [2003](#), MSc Thesis Edberg [2007](#), PhD Thesis Grace [2004](#) or articles within conference proceedings Park, Kim, and Jeon [2006](#). Several other types of article exist, but they are used to a lesser degree.

To change between referencing styles, go to line 21 of `thesis.tex` where it says `%%% REFERENCING STYLES %%%`, and comment/uncomment accordingly! The following subsections will explain more about both styles (NOTE: This pdf has been compiled using the Harvard referencing style, as it is the one preferred by the University). More information about referencing according to RGU guidelines can be found here:

1. <https://library.rgu.ac.uk/referencing-updates-staff>
2. <https://library.rgu.ac.uk/referencing-and-refworks>

7.4.1 The Harvard Bibliography Style

The bibliographic references in this document laid out using the Harvard style. Further information about this style may be found at the following links:

1. <https://library.rgu.ac.uk/RGUPharvard>
2. <https://library.rgu.ac.uk/harvard-referencing-templates>
3. <https://library.rgu.ac.uk/ctr-harvard-referencing-templates>

If you want parenthesis to enclose the reference, use *parencite* or *autocite* instead of *cite*. This would change the citation style in the previous paragraph from:

Einstein [1956](#) for example is well known for his work on Brownian Movement.

TO

(Einstein [1956](#)) for example is well known for his work on Brownian Movement.

7.4.2 The Vancouver Bibliography Style

Most readers recommend using the Vancouver citation style, as it enhances readability and allows you to refer to papers through a number within squared brackets [x] rather than using last names and years within the text, which can sometimes be confusing. From 1st September 2024 onwards, RGU has decided also to allow this style as mentioned in the following links:

1. <https://library.rgu.ac.uk/ctr-vancouver-referencing-templates>

7.4.3 Compiling a BibTeX Database

Having initially compiled the document using pdfLatex a number of helper files are created that aid in referencing and citations. One then must compile the bibtex database, followed by an additional two compiles using pdfLatex. Citing additional bibliographic references within the body of the document being produced will require the recompile of the bibtex database. In the case that the bibtex reference of a cited article cannot be found one will see **a text in bold** (in case of using Harvard) or a question mark inside squared brackets [?] (in case of using Vancouver) instead of the proper citation.

7.5 Working with Tables

The data in a table (Table 7.1) displays four columns of left-aligned data. The cell contents can be aligned to the left (l), right (r), or center (c). Vertical bars may sometimes be seen in tables but these generally look unprofessional. The Booktabs package Fear 2005 allows for the creation of professional looking tables as shown in the example. The use of `\toprule`, `\midrule` and `\bottomrule` commands provided by the package allow for rules of varying thickness and spacing. Data elements (cells / columns) within a table are divided up using the ampersand (&). To complete a row one must end with a double backslash (`\`). Tables as with figures need a caption and a label.

Heading 1	Heading 2	Heading 3	Heading 4
Cell A1	Cell B2	Cell C3	Cell D4
Cell E1	Cell F2	Cell G3	Cell H4

Table 7.1: Table Caption

One may find the following spreadsheet tool (<http://cobweb.ecn.purdue.edu/~zhang97/xls2latex/>) to be particularly useful for quickly converting tabular data

in a spreadsheet to L^AT_EX form. A complete list of instructions on how to use the tool are also present. The WinShell editor has an in-built GUI based utility to aid in the creation of the tabular data.

7.6 Inserting an Algorithm

The *algorithm2e* environment may be used to generate algorithms (Algorithm 1). The following document (<http://www.tex.ac.uk/ctan/macros/latex/contrib/algorithm2e/algorithm2e.pdf>) provides detail of all the commands available within the package. If no algorithms are used within the document then comment out line 70 of the file `thesis.tex` to remove the list of algorithms from the contents area.

Algorithm 1: A Sample Algorithm

```

while (RANK < COMPSIZE) do
|   if (RANK == MASTER) then
|   |   generate random value ;
|   |   for (each item K) do
|   |   |   get result ;
|   |   end
|   end
end

```

7.7 Inserting Program Code Samples

To insert small segments of program code (Listing 7.2) that detail how interesting algorithms and so forth are implemented use the *lstlisting* command. Inclusion of program code again requires a Caption and Label. Sample code from external files may also be included (Listing 7.1), again one must supply a Caption and a Label as well as the relative path to the source file.

```

1 public void someInterestingMethod( int x ) {
2     try {
3         DataInputStream in = new DataInputStream(req.getInputStream());
4         HttpSession session = req.getSession(true);
5         //Read Input Data
6         in.close();
7     } catch ( Exception err ) { }
8     outputStream.close();
9 }

```

Listing 7.1: The Caption for the Code Listing

Alternatively, you can declare it using the following notation.

```

1  if(rndVal==0){
2      if(opType > 2){
3          //do something
4      }
5  }

```

Listing 7.2: Sample Program Code Listing

7.8 Working with Maths

One can insert mathematical formula directly into a paragraph of text. The mathematical definition of the “Cantor set” is a good example of this in action $\sum_{n=0}^{\infty} \frac{2^n}{3^{n+1}} = \frac{1}{3} + \frac{2}{9} + \frac{4}{27} + \frac{8}{81} + \dots = \frac{1}{3} \left(\frac{1}{1 - \frac{2}{3}} \right) = 1$. The previous equation demonstrates the use of sigma, fractions, large brackets, power, and dots. The function that defines the MSet is a simpler example of math in use $Z_{n+1} = Z_n^2 + C$. Matrix Multiplication is typically regarded as an $O(n^3)$ operation. One may use the *equation* environment for more complex mathematical formula that should stand out. One may take for example the product C of two matrices $A \in M_{n,m}(R)$ and $B \in M_{m,p}(R)$ to be defined as

$$(A \times B)_{ij} = \sum_{k=0}^{m-1} a_{ik} b_{kj}, \quad i = 0, \dots, n-1, \quad j = 0, \dots, p-1. \quad (7.1)$$

The sizes of the matrices must satisfy $(n \times m)(m \times p) = (n \times p)$. Matrix multiplication is an associative process thereby $a \cdot (b \cdot c) = (a \cdot b) \cdot c$. Essentially to find the value of a particular cell $C_{i,j}$ it is necessary to multiply row i of the matrix A with column j of matrix B summing all the multiplications.

7.9 Required Software

The implementation of L^AT_EX typically used is MikTeX (<http://miktex.org/>). It is best to install the complete MiKTeX system. Initially a small installer application must be downloaded and executed. This in-turn downloads the most recent implementation of the MikTeX system. The complete system is circa 500MB in size. Run the installer again and select the directory of the downloaded package.

To allow one to work with postscript documents a system called Ghostscript is required (<http://pages.cs.wisc.edu/~ghost/>). To be able to view these documents download and install GhostView (<http://pages.cs.wisc.edu/~ghost/>

gsview/index.htm). The last element that is necessary is an editor. TeXnicCenter is a free download available at (<http://sourceforge.net/projects/texniccenter/>). An alternative is WinEdt a shareware ASCII editor (<http://www.winedt.com/>). WinEdt can be freely used for a 30 day period, after which one will periodically receive reminders to register the product. To demonstrate the use of an enumerated list, the software should be installed in the order detailed below. One final option for Windows users is WinShell (<http://www.winshell.de/>). An advantage of WinShell is its in-built BibTeX GUI editor. It also features a useful Table Wizard.

1. MikTeX
2. Ghostscript
3. Ghostview
4. Editor

7.10 Working with Quotes

To surround a piece of text with double quotes one must place two single quotes on either side of the text. The double quote on the left is created using two left quotes (‘) this is located just above the *tab* key on the keyboard. The right hand double quote is created using two right hand quotes (’) located just above and to the left of the right shift key. A properly formatted quotation should look like “This is a quotation”. Notice how the direction of the quotes are opposite to one another.

7.11 Further Information

The Not so Short Introduction to LaTeX (<http://tobi.oetiker.ch/lshort/lshort.pdf>) is one useful source of further information on how to work with this system.

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Appendix A

Project Specification

Summary of the project outline.

A.1 Functional Requirements

some text here

A.2 Non-Functional Requirements

some text here

Appendix B

Project Management

Discussion on how the project was managed. What things impacted the success of the project. How does the continually revised versions of the project plan compare to the initial draft developed at the start of the project. Did everything run according the schedule. Did elements such as exams & coursework have any impact.

Appendix C

Another Appendix

This appendix makes use of the *rotating* package to rotate both figures and tables ninety degrees allowing for large datasets and illustrations to be represented.

Heading 1	Heading 2	Heading 3	Heading 4	Heading 5	Heading 6	Heading 7	Heading 8	Heading 9	Heading 10
aaa	bbbb	ccc	dddd	eeee	fff	gggg	hhhh	iiii	jjjj
aaa	bbbb	ccc	dddd	eeee	fff	gggg	hhhh	iiii	jjjj
aaa	bbbb	ccc	dddd	eeee	fff	gggg	hhhh	iiii	jjjj
aaa	bbbb	ccc	dddd	eeee	fff	gggg	hhhh	iiii	jjjj
aaa	bbbb	ccc	dddd	eeee	fff	gggg	hhhh	iiii	jjjj
aaa	bbbb	ccc	dddd	eeee	fff	gggg	hhhh	iiii	jjjj

Table C.1: A much longer caption that will not be listed in the list of tables page.



Figure C.1: A much longer caption that will not be listed in the list of figures page.

Appendix D

Presentation Slides

The slides from the formal presentation should be provided here in not more than two pages.

Appendix E

Project Log

The following is a weekly summary of the work carried during the development of this body of work. It covers tasks that were completed, tutorials that were worked through, articles that were read and reviews of discussions / meetings held with the project supervisor and other third parties.

Week Beginning: Monday 27/09/2010

First week working on the project. Had a meeting with supervisor and discussed some of the issues related to the project. The first deliverable is due for the end of next week (project outline & ethics form).

- Downloaded and Installed L^AT_EX (MikTeX full install), Ghostscript, Ghostview & Winshell.
- Started to get to grips with the L^AT_EX system by making simple modifications to the template and editing the project log.
- Developed a Mind Map to clarify understanding of project elements.
- Prepared an initial draft of project plan in the form of a Gantt chart.
- Prepared and revised 1 page draft of project summary & filled in ethics form.